

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run CNN-LSTM Attention Model
2. Abbreviated Title	Run Model Inference
3. Use Case Id	UC-11
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine runs forward inference through the hybrid CNN-LSTM-Attention architecture using preprocessed inputs (normalized vector + GAF image) to produce a binary heart disease risk probability score.
5.1. Pre Conditions	- Preprocessed data is available from UC-10. - Trained model checkpoint is loaded into memory.
5.2. Task Sequence	- CNN branch processes the GAF image to extract spatial features. - LSTM branch processes the feature sequence for sequential patterns. - Attention mechanism computes weighted feature importance. - Fused representation passes through the classification head. - System outputs sigmoid probability and risk label (HIGH/LOW).
5.3. Post Conditions	- Binary classification output (probability + label) is produced. - Internal activations are stored for XAI computation.
6. Modification History	Date: September 11, 2026
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